

Assignment 2

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Course : SECR1033 (Computer Organization
Detail : And Architecture)

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Q1

(i) ESI = 40000000h

(ii) EDI = 40000003h

(iii) CL = 0Ah

(iv) BX = 1E14h

(v) EDX = 0001F403h

(vi) BX = 01F4h

40000000h	0A	my bytes
1h	14	my bytes + 1
2	1E	+ 2
3	E8	my words
4	03	my words + 1
5	F4	+ 2
6	01	+ 3
7	00	+ 4
8	00	+ 5

Question 2

a)

3000h	69
3001h	00
3002h	0B
3003h	00
3004h	6A
3005h	00

} Set Arr 1

3000h	
3006h	DF
3007h	
3008h	5A
3009h	43 5A
300Ah	43
300Bh	41
300Ch	52

} Set Arr 2

b) (i) 3002h = 000Bh

Ax = 000Bh

(iii) 3003h = 00

3004h = 6A

Bx = 6A00h

(ii) ~~3000h~~ 3009h = 5Ah

DH = 5Ah

(iv) 300Ah = 43h

DL = 43h

c) Yes, the program will still assemble and run without error. because Dumpregs is only used for displaying register value and not required for program execution.

$$Q_3) Q = ((A+D) * (C-D)) / (E + F * G)$$

~~AX, AB, AD~~

a) ADD T1, A, B
 MPY T1, T1, C
 SUB T1, T1, D
 MPY T2, F, G
 ADD T2, E, T2
 DIV Q, T1, T2

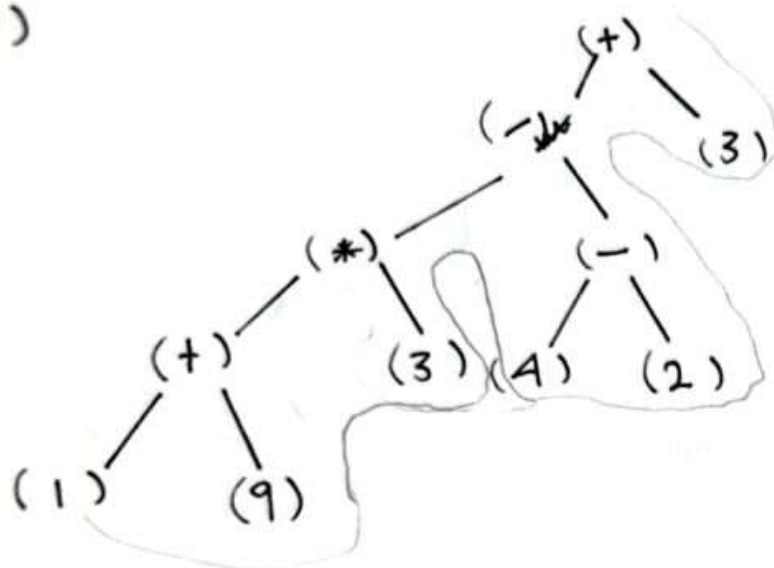
b) MOVE T1, A
 ADD T1, B
 MPY T1, C
 SUB T1, D
 MOVE T2, F
 MPY T2, G
 ADD T2, E
 DIV T1, T2
 MOVE Q, T1

c) LOAD F
 MPY G
 ADD E
 STOR T1
 LOAD A
 ADD B
 MPY C
 SUB D
 DIV T1
 STOR Q

d) PUSH A
 PUSH B
 ADD
 PUSH C
 MUL
 PUSH D
~~MPY~~ SUB
 PUSH E
 PUSH F
 PUSH G
 MUL
 ADD

Question 4

a)



b) RPN = 19+3*42-/3+

1. $1+9=10$

2. $10 \times 3=30$

3. $4-2=2$

4. $30/2=15$

5. $15+3=18$

$M=18$

c) (i) Load 1
Add 9
MUL 3
STORE T1

Load 4
SUB 2
STORE T2

LOAD T1
DIV T2
ADD 3
STORE H

(ii) PUSH 1
PUSH 9
ADD
PUSH 3
MUL
PUSH 4
PUSH 2
SUB
DIV
PUSH 3
ADD
POP M